Repeat Purchase Prediction

Homework – Overall Project Structure and Rationale

Binary classification of whether a customer purchases again within 90 days

NOVA IMS DSAA Master's 2026-2027

Machine Learning

Group 11

João André Mateus Neves 20241487@novaims.unl.pt

Tomas Castanheira Muacho 20260678@novaims.unl.pt

Fabiana Maria Carreiro Rebelo 20260540@novaims.unl.pt

Flávia Daniela Motta 20230574@novaims.unl.pt

1. Objective and design principle

The task is to predict **repeat_purchase_90d** for each purchase occasion. The workflow was designed around two principles: each modelling decision should follow from the exploratory analysis, and every transformation that learns from data must be fitted only on training observations. This keeps the process reproducible and limits data leakage when performance is estimated.

2. Pipeline overview

1. Explore Structure, target, missingness, anomalies, distributions and relationships	**2. Clean & engineer** Remove empty rows, correct impossible values, create first-purchase/date features	**3. Preprocess** Impute, cap extremes, encode categories and scale numeric variables
4. Select features Filter weak/redundant variables, then use L1 regularisation	**5. Validate model** Logistic regression assessed with 5 chronological folds using F1	**6. Final prediction** Refit on all training rows and generate test predictions in submission format

3. Exploration and data preparation

Exploration showed that the data are purchase occasions rather than independent customers, that customers can appear repeatedly, and that the test period occurs after the training period.

The cleaned training set contains **6,469 rows** after removing 65 rows with no predictor information; the test set contains **986 rows**. The training target rate is about **56%**.

Data-quality checks identified impossible negative values and sentinel values such as 1,000,000,000. These were converted to missing values rather than treated as genuine observations.

Missing historical variables were often structural: a customer making a first purchase cannot have previous-purchase history. An **is_first_purchase** indicator was therefore created before imputation.

Dates were parsed and month/day-of-week features were created. **ID** and **customer_id** were retained for joins and submission but excluded from the estimator.

The optional customer table was assessed separately; only **commercial_region** was initially added, and its value was later tested rather than assumed.

4. Reproducible preprocessing

Numeric gaps are median-imputed with missing-value indicators, categorical gaps receive an explicit missing category, rare/unseen categories are handled during one-hot encoding, and numeric variables are scaled.

Extreme but plausible numeric values are not deleted; instead, the chosen model caps them at the **0.5% and 99.5% quantiles**, with those limits learned inside each training fold.

All preprocessing is contained in a scikit-learn pipeline, so validation observations never determine medians, category thresholds, caps or scaling parameters.

5. Feature selection rationale

Feature selection combines an exploration-based filter with an embedded method.

First, variables with no useful signal, near-constant behaviour or strong redundancy were removed only when cross-validation showed no meaningful loss in F1. This removed **17 original columns**, including random_noise, weak categorical/numeric variables, redundant invoice totals and overlapping purchase-history measures. The resulting model retained **22 of 39 candidate original features**.

Second, logistic regression with an **L1 penalty** was used as an embedded selector. L1 sets some encoded coefficients exactly to zero and therefore reduces the effective feature set while training the model. Among the tested values, **C = 0.2** produced the strongest mean F1 improvement.

The largest coefficient was associated with **purchase_count_365d**, consistent with the exploratory finding that recent purchase frequency is a strong repeat-purchase signal.

6. Model assessment

Because the final test observations occur later in time, random validation would not reproduce the intended prediction setting. The training data are therefore sorted by purchase date and evaluated with **TimeSeriesSplit using five chronological folds**: each fold learns from earlier purchases and validates on the period immediately after them. This also shows how stable performance is across different time periods.

F1 is the primary metric because it is the competition metric and is therefore used consistently for model decisions. **ROC AUC** is reported as a secondary measure of ranking quality. An 'always predict 1' rule is also included as a minimum reference because the positive class is the majority.

Model stage	Mean F1	Mean AUC
Always predict 1	0.721	–
Baseline logistic regression	0.733	0.783
+ 0.5% / 99.5% capping	0.737	0.784
+ filter selection	0.739	0.782
+ L1 penalty (C = 0.2)	**0.750**	**0.791**

7. Final model and current result

The current “homework model”, for this particular mid-semester submission, is a **logistic regression with L1 regularisation (C = 0.2)**, using the selected features and the preprocessing choices above.

Across the five chronological folds it reaches approximately **F1 = 0.750 ± 0.026** and **AUC = 0.791**. It is then refitted on all available training observations and used to predict all 986 test rows while preserving the original ID order required by the sample submission.

The resulting Kaggle submission achieved a score of **0.7896**.

This model is intentionally simple: its purpose is to establish a transparent, reproducible benchmark for the mid-semester homework/deliverable.

The full project can subsequently compare additional classification algorithms, tune hyperparameters and decision thresholds, and test whether more complex approaches provide a reliable improvement.

8. Why this structure is appropriate

- **Traceable:** cleaning and feature decisions are linked to findings from exploration.
- **Leakage-aware:** learned preprocessing is fitted inside each chronological training fold.
- **Comparable:** preprocessing and feature-selection alternatives are evaluated with the same folds and F1 metric.
- **Parsimonious:** the final homework model is deliberately simple and interpretable while still improving on the baseline.
- **Submission-ready:** the pipeline ends with predictions for every test observation and preserves ID for the required Kaggle format.